

SHETH L.U.J AND SIR M.V COLLEGE
SUBJECT :- CYBER SECURITY
PRACTICAL NO:- 3

AIM:- Implement algorithms to generate and verify message authentication codes (MACs) for ensuring data integrity and authenticity.

INPUT:-

```
import hmac
import hashlib

# Secret key shared between sender and receiver
secret_key = b"mysecretkey"

# Message
message = input("Enter the message: ").encode()

# Generate MAC
mac = hmac.new(secret_key, message, hashlib.sha256).hexdigest()

print("\nGenerated MAC:", mac)

# Verification
received_message = input("\nEnter received message: ").encode()
received_mac = input("Enter received MAC: ")

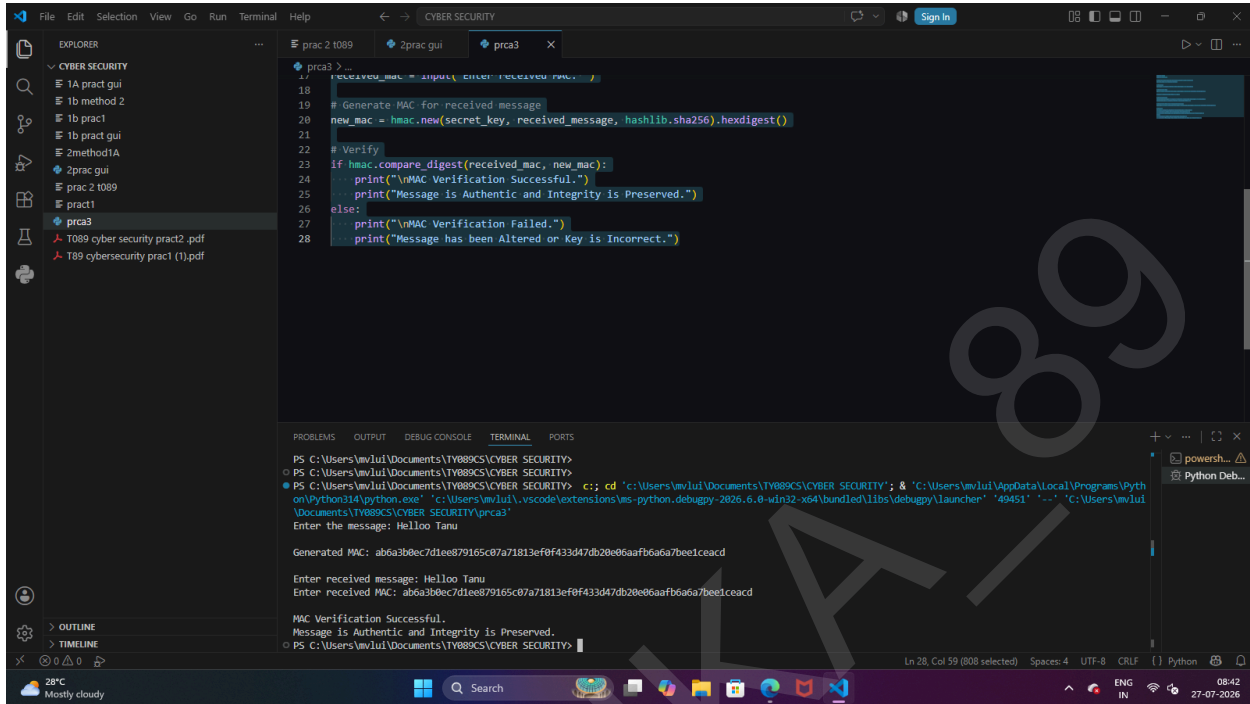
# Generate MAC for received message
new_mac = hmac.new(secret_key, received_message,
hashlib.sha256).hexdigest()

# Verify
if hmac.compare_digest(received_mac, new_mac):
    print("\nMAC Verification Successful.")
    print("Message is Authentic and Integrity is Preserved.")
else:
    print("\nMAC Verification Failed.")
    print("Message has been Altered or Key is Incorrect.")
```

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OUTPUT:-



The screenshot shows a VS Code editor with a file explorer on the left containing files like '1A pract gui', '1b method 2', '1b pract1', '2method1A', '2prac gui', 'prac 2 1089', 'prac1', and 'prac3'. The main editor window displays a Python script named 'prca3.py' with the following code:

```
17 received_mac = input("Enter received MAC: ")
18
19 # Generate MAC for received message
20 new_mac = hmac.new(secret_key, received_message, hashlib.sha256).hexdigest()
21
22 # Verify
23 if hmac.compare_digest(received_mac, new_mac):
24     print("\nMAC Verification Successful.")
25     print("Message is Authentic and Integrity is Preserved.")
26 else:
27     print("\nMAC Verification Failed.")
28     print("Message has been Altered or Key is Incorrect.")
```

The terminal output at the bottom shows the execution of the script:

```
PS C:\Users\mv\lul\Documents\TY089CS\CYBER SECURITY>
PS C:\Users\mv\lul\Documents\TY089CS\CYBER SECURITY> cd "c:\Users\mv\lul\Documents\TY089CS\CYBER SECURITY"; & "C:\Users\mv\lul\AppData\Local\Programs\Python\Python314\python.exe" "c:\Users\mv\lul\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher" "49451" "-" "C:\Users\mv\lul\Documents\TY089CS\CYBER SECURITY\prca3"
Enter the message: Hello Tanu

Generated MAC: ab6a3b0ec7d1ee879165c07a71813ef0f433d47db20e06aafb6a6a7bee1ceacd
Enter received message: Hello Tanu
Enter received MAC: ab6a3b0ec7d1ee879165c07a71813ef0f433d47db20e06aafb6a6a7bee1ceacd

MAC Verification Successful.
Message is Authentic and Integrity is Preserved.
```

GUI:-



The screenshot shows a Python script for a GUI-based MAC generator. The code imports tkinter, messagebox, hmac, and hashlib. It defines a SECRET_KEY and a generate_mac function that takes a message from a text entry and returns a MAC. The function also includes a warning message if the message is empty.

```
import tkinter as tk
from tkinter import messagebox
import hmac
import hashlib

# Secret Key
SECRET_KEY = b"mysecretkey"

# Generate MAC
def generate_mac():
    message = entry_message.get()

    if message == "":
        messagebox.showwarning("Warning", "Please enter a message!")
        return

    mac = hmac.new(SECRET_KEY, message.encode(),
hashlib.sha256).hexdigest()
```

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```
entry_mac.delete(0, tk.END)
entry_mac.insert(0, mac)

# Verify MAC
def verify_mac():
    message = entry_verify_message.get()
    received_mac = entry_verify_mac.get()

    if message == "" or received_mac == "":
        messagebox.showwarning("Warning", "Please enter message and MAC!")
        return

    new_mac = hmac.new(SECRET_KEY, message.encode(),
hashlib.sha256).hexdigest()

    if hmac.compare_digest(new_mac, received_mac):
        lbl_result.config(
            text="✓ MAC Verified\nMessage is Authentic",
            fg="green"
        )
    else:
        lbl_result.config(
            text="✗ MAC Verification Failed\nMessage Modified",
            fg="red"
        )

# ----- GUI ----- #

root = tk.Tk()
root.title("Message Authentication Code (MAC)")
root.geometry("600x430")
root.resizable(False, False)
root.configure(bg="#E8F6F3")

title = tk.Label(
    root,
    text="Message Authentication Code (HMAC-SHA256)",
    font=("Arial", 16, "bold"),
    bg="#E8F6F3",
```

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```
        fg="navy"
    )
    title.pack(pady=10)

    # Generate Section
    frame1 = tk.LabelFrame(root, text="Generate MAC", font=("Arial", 11,
    "bold"), bg="#E8F6F3")
    frame1.pack(fill="x", padx=15, pady=10)

    tk.Label(frame1, text="Message:", bg="#E8F6F3").grid(row=0, column=0,
    padx=10, pady=10)

    entry_message = tk.Entry(frame1, width=55)
    entry_message.grid(row=0, column=1, padx=10)

    tk.Button(
        frame1,
        text="Generate MAC",
        bg="#3498DB",
        fg="white",
        command=generate_mac
    ).grid(row=1, column=1, pady=10)

    tk.Label(frame1, text="Generated MAC:", bg="#E8F6F3").grid(row=2,
    column=0, padx=10, pady=10)

    entry_mac = tk.Entry(frame1, width=55)
    entry_mac.grid(row=2, column=1)

    # Verify Section
    frame2 = tk.LabelFrame(root, text="Verify MAC", font=("Arial", 11,
    "bold"), bg="#E8F6F3")
    frame2.pack(fill="x", padx=15, pady=10)

    tk.Label(frame2, text="Received Message:", bg="#E8F6F3").grid(row=0,
    column=0, padx=10, pady=8)

    entry_verify_message = tk.Entry(frame2, width=55)
    entry_verify_message.grid(row=0, column=1)
```

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```
tk.Label(frame2, text="Received MAC:", bg="#E8F6F3").grid(row=1, column=0,
padx=10, pady=8)

entry_verify_mac = tk.Entry(frame2, width=55)
entry_verify_mac.grid(row=1, column=1)

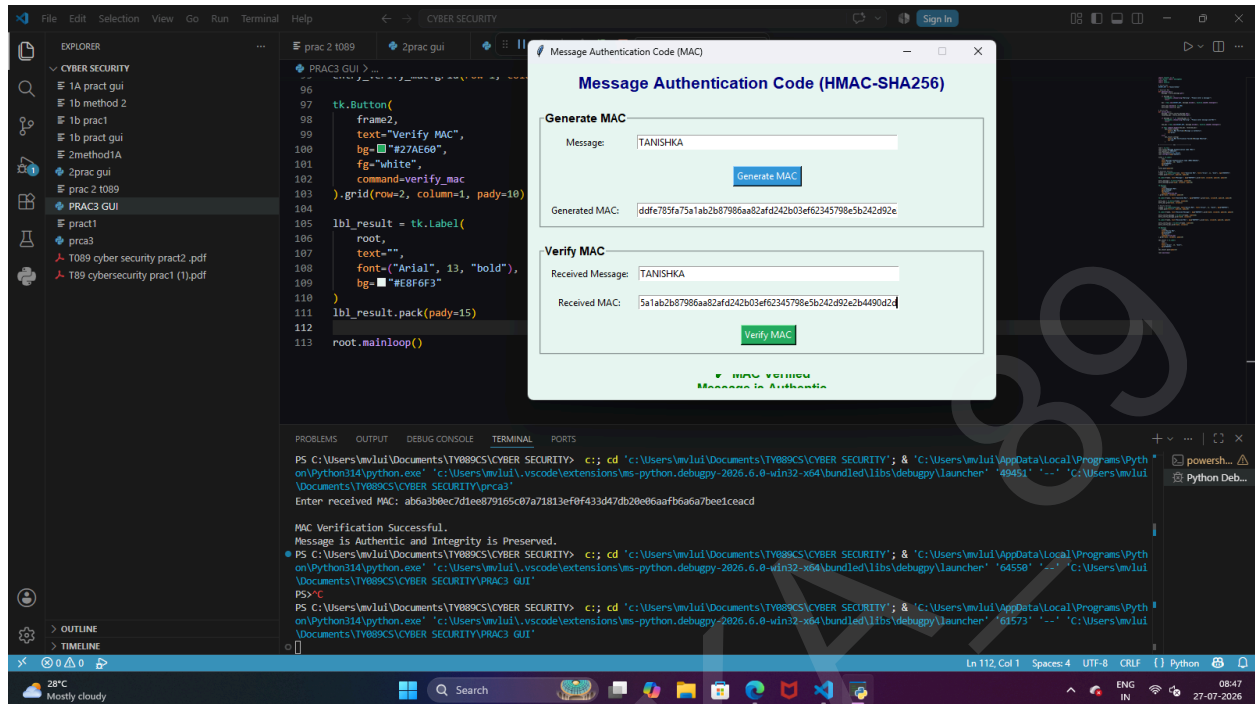
tk.Button(
    frame2,
    text="Verify MAC",
    bg="#27AE60",
    fg="white",
    command=verify_mac
).grid(row=2, column=1, pady=10)

lbl_result = tk.Label(
    root,
    text="",
    font=("Arial", 13, "bold"),
    bg="#E8F6F3"
)
lbl_result.pack(pady=15)

root.mainloop()
```

OUTPUT:-

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